Manufacturing Technology Project Report: 3D Printed Handheld Fan

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Abstract

This report presents the design and additive manufacture of a hand-cranked, gear-driven handheld fan with no motor or battery. The product is an assembly of two shell halves, a crank handle, a large spur gear, a pinion shaft, a propeller, a propeller cap and axle pins. It was modelled in CAD and checked with an exploded-view assembly study. The project aims to show that Fused Deposition Modelling (FDM) can produce a working multi-part mechanism without tooling, using a single material and no electronics. The report covers design requirements, process and material selection, slicing strategy, post-processing, assembly, testing and cost. The key finding is that a speed-increasing gear stage lets a slow crank turn the propeller fast enough to give a usable airflow, while every part can be printed without supports in a flat orientation. The main limits are gear tooth wear and layer-line friction, which post-processing reduces. Print settings and costs are typical FDM values and should be replaced with the team's measured figures.

1. Introduction

A hand-cranked fan is a good test of whether a 3D printer can make a real machine, not just a shape. It needs accurate gear meshing, rotating joints and a balanced rotor, which are the same demands placed on much larger products.

1.1 Motivation

Portable fans normally use injection-moulded housings and electric motors. Moulds cost tens of thousands of rupees and only pay off at high volume. Additive manufacturing removes the mould, so a single custom unit costs the same per part as a hundred. A manual drive also needs no battery, works in power cuts and is easy to repair.

1.2 Objectives

- Design a compact handheld fan driven by a hand crank and a gear train.
- Select an FDM material and print settings that suit gears, shafts and thin shells.

- Produce all parts without support material, or with minimal supports.
- Assemble and verify the mechanism using a CAD exploded-view study.
- Estimate manufacturing time and cost, and list the defects to expect.

1.3 Scope and assumptions

The video supplied shows the CAD model and its exploded view only. Dimensions, tooth counts, filament and printer are not shown, so the figures in sections 5 to 9 are typical values for desktop FDM and are marked as assumed. Replace them with measured data before submission.

2. Background: Additive Manufacturing and FDM

Additive manufacturing (AM) builds a part layer by layer from a digital model, in contrast to subtractive methods such as milling that cut material away. ISO/ASTM 52900 groups AM into seven process families. FDM, also called material extrusion, is the most common for education and prototyping.

2.1 How FDM works

A spool of thermoplastic filament is pushed into a heated nozzle, melted and deposited along a path set by the slicer. The bed or head moves down one layer height and the next layer fuses to the one below. Typical layer heights are 0.1 to 0.3 mm and nozzle diameters 0.4 mm.

2.2 Why FDM suits this product

Requirement	FDM capability
Low cost, single units	No tooling; cost depends on material and machine time
Moving parts	Tolerances of about ±0.2 mm allow clearance fits
Complex shapes	Curved shell and propeller blades need no mould
Strength	Adequate for a lightly loaded hand-driven gear train

2.3 Limitations to design around

- Parts are weaker across layers than along them (anisotropy).
- Overhangs steeper than about 45° need supports or redesign.
- Surfaces show layer lines, which add friction at bearings and gear flanks.
- Plastics creep and wear faster than metal under sustained load.

3. Product Design

3.1 Functional description

The user holds the stadium-shaped body and turns the crank handle. The crank drives a large spur gear, which meshes with a small pinion on the propeller shaft. Because the pinion has fewer teeth, the propeller turns several times faster than the crank. The propeller sits outside the body at one end, held on by a cap.

3.2 Components seen in the CAD assembly

Component	Function	Qty
Shell half (front, with pin bosses)	Holds the gear train and forms the grip	1
Shell half (back, with locating pins)	Closes the body and aligns the halves	1
Crank handle	Input; converts hand motion to rotation	1
Large spur gear	First stage of the speed-up	1
Pinion and propeller shaft	Second stage; carries the propeller	1
Propeller	Produces the airflow	1
Propeller cap	Retains the propeller on the shaft	1
Axle pins	Support the gear and crank shaft inside the body	2 to 3

3.3 Design requirements

- Overall body small enough to hold in one hand.
- Crank must turn smoothly at 1 to 2 revolutions per second.

- Propeller must clear the body and spin freely.
- All parts printable on a standard 220 × 220 mm bed.

3.4 Gear ratio

The speed ratio of a spur pair is set by the tooth counts:

$$i = \frac{N_{gear}}{N_{pinion}}, \qquad n_{prop} = i \cdot n_{crank}$$

For example, a 60-tooth gear driving a 12-tooth pinion gives i = 5, so a crank at 90 rpm drives the propeller at 450 rpm. The tooth counts here are illustrative; take the real values from the CAD model.

3.5 CAD and assembly study

The model was assembled and an exploded view generated with the CAD system's Explode tool, using a scroll slider from 0 (assembled) to 1 (fully exploded). The study confirmed that every part has a clear assembly direction: shafts slide in along their axes, the two shell halves close over the gear train, and the cap is the last part fitted.

4. Material Selection

PLA is the recommended material for the whole assembly. It prints reliably, warps little and gives the sharpest gear teeth. PETG and nylon are better options if the fan must survive heat or heavy use.

Property (typical)	PLA	PETG	ABS	Nylon
Print temperature (°C)	190 to 220	220 to 250	230 to 260	240 to 270
Tensile strength (MPa)	50 to 60	45 to 50	35 to 45	40 to 70
Warping	Low	Low	High	High
Wear resistance	Fair	Good	Good	Excellent
Ease of printing	Easiest	Easy	Hard	Hard

Values are approximate manufacturer ranges, not measurements from this project.

4.1 Rationale

- **Shells and crank:** PLA gives stiff, good-looking parts with low warp over a long body.
- **Gears and pinion:** PLA is acceptable for a hand-driven fan. Nylon would last longer if the gears show wear.
- **Pins:** Printed pins are simplest. A steel pin or 3 mm rod can replace them where a shaft carries more load.
- **Colour:** The CAD model is blue throughout, which suggests one colour and one material for a clean build.

5. Manufacturing Process

The parts are made by FDM in the sequence below. All settings are typical starting values (assumed) for a 0.4 mm nozzle and PLA.

1. Export each part from CAD as an STL file at fine tessellation so gear teeth stay smooth.
2. Check the mesh for holes and flipped faces, then import into the slicer.
3. Orient each part (section 5.1) and set the print parameters (section 5.2).
4. Slice, preview layer by layer, and export G-code.
5. Print, monitoring the first layer and removing parts once the bed has cooled.
6. Post-process (section 5.3) and inspect.

5.1 Build orientation

Part	Orientation	Reason
Shell halves	Flat, open side up	Large flat face on the bed; bosses and pins grow upward without supports
Large gear	Flat	Teeth are cut in the XY plane, so they are strongest and most accurate
Pinion and shaft	Shaft upright or flat	Upright gives a round shaft; flat gives stronger layers along its length
Crank handle	Flat	Bending load acts along the layers
Propeller	Flat, hub down	Blades are thin and angled; low supports or a brim keep them attached

Part	Orientation	Reason
Cap and pins	Flat or upright	Small parts; brim or raft helps adhesion

5.2 Print parameters (assumed)

Parameter	Shells	Gears and shafts	Propeller
Layer height (mm)	0.20	0.12 to 0.16	0.12
Walls	3	4	3
Infill (%)	20	50 to 100	30
Nozzle (°C)	210	210	210
Bed (°C)	60	60	60
Speed (mm/s)	50	30 to 40	30
Supports	None	None	Only if blades need them

Fine layers and slow speeds on the gears improve tooth shape and reduce noise. Extra walls matter more than infill for gear and shaft strength.

5.3 Post-processing

- Remove brims and stringing with a hobby knife.
- Sand gear flanks and shafts lightly (400 to 800 grit) to remove layer ridges.
- Ream pin holes with a drill bit sized to the pin so shafts spin freely.
- Lubricate gear teeth and shafts with a drop of silicone grease or machine oil.
- Balance the propeller by checking that it stops at random angles when spun on its shaft.

6. Assembly and Mechanism

There is no electronics in this design. Assembly is purely mechanical, and the exploded view in the CAD study gives the order directly.

1. Place the large gear and its crank shaft on the pins in the open shell half.

2. Fit the pinion and propeller shaft so its teeth mesh with the large gear. Turn by hand to check there is no binding.
3. Close the second shell half so its locating pins enter the bosses of the first.
4. Slide the crank handle onto its shaft and secure it.
5. Fit the propeller on the protruding shaft and fix it with the cap.
6. Spin the crank 10 turns without load to bed the gears in.

6.1 Fits and clearances (assumed)

Feature	Clearance (mm)	Note
Rotating shaft in hole	0.2 to 0.3	Free spin; ream if tight
Locating pin in boss	0.1 to 0.2	Snug, aligns shell halves
Gear centre distance	+0.1 to 0.2	Prevents tooth jamming
Cap on shaft	0.0 to 0.1	Light press fit

Printed parts expand slightly at hole edges, so holes are modelled larger than the shaft they carry. The values come from common FDM practice and should be tuned on a test print.

7. Testing, Defects and Observations

The video verifies the design in CAD only, so the physical test results are not available. The plan below lists what to measure and the defects to expect. Fill the results column with your own readings.

7.1 Test plan

Test	Method	Pass criterion	Result
Fit check	Assemble without force	Parts mate; shells close fully	
Free rotation	Turn crank by hand	No binding over a full turn	
Speed ratio	Count propeller turns per 10 crank turns	Matches tooth-count ratio	

Test	Method	Pass criterion	Result
Airflow	Handheld anemometer 20 cm in front	Noticeable airflow, e.g. above 1 m/s	
Endurance	Crank continuously for 5 minutes	No tooth skipping or cracks	
Balance	Spin propeller and observe vibration	No visible wobble	

7.2 Expected defects and remedies

Defect	Cause	Remedy
Gears bind or click	Tight centre distance, layer ridges	Increase clearance, sand flanks, lubricate
Warped shell	Thermal contraction on a long part	Heated bed, brim, draught shield
Cracked crank	Weak layer bonding across the load	Print flat, add walls, raise nozzle temperature
Propeller wobble	Uneven blades or loose hub	Reprint finer, sand to balance, tighten fit
Stringing	Moisture or high temperature	Dry filament, lower temperature, retraction

7.3 Observations from the CAD study

The exploded view shows that the shell halves are symmetric enough to print without supports and that the gear train is accessible from one side, which makes the printed prototype easy to open and fix. The propeller sits outside the shell, so its blade clearance can be checked visually.

8. Cost and Time Analysis

The estimate is about ₹150 to ₹250 in material and 8 to 12 hours of print time per fan. All inputs are assumed typical values for desktop FDM in India (PLA at roughly ₹1,000 per kg); replace them with your own.

Item	Assumed mass (g)	Assumed print time (h)	Cost (₹)
Shell halves (2)	70	5.0	70
Large gear	12	1.0	12
Pinion and shaft	5	0.5	5
Crank handle	8	0.7	8
Propeller and cap	10	1.0	10
Pins	3	0.3	3
Total material	108	8.5	108
Failed prints and supports (allowance 30%)			32
Lubricant and sandpaper			30
Total, excluding machine time			**170**

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C_{unit} = C_{filament} + C_{waste} + C_{consumables} + t_{print} \cdot C_{machine}
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At ₹30 per machine hour, 8.5 hours adds about ₹255. An injection-moulded equivalent would cost a few rupees per unit but needs a mould costing lakhs, so AM wins below roughly a few thousand units.

9. Conclusion

A hand-cranked fan with a gear train, crank, propeller and two-part shell can be made entirely by FDM from one material, with no supports and no electronics. The exploded-view study confirms a clean assembly sequence.

9.1 Key takeaways

- Print gears and shafts flat, with fine layers and extra walls, for accuracy and strength.
- Clearances of 0.2 to 0.3 mm on rotating fits are what make a printed mechanism run.
- Sanding and lubrication matter as much as print settings for smooth cranking.

- AM is the right route for one-off or low-volume units, where moulds are uneconomic.

9.2 Limitations

PLA gears wear and creep, thin propeller blades are fragile, and layer lines add friction. The data in this report beyond the CAD model are assumed and need measuring.

9.3 Future work

- Use nylon or PETG gears, or metal shafts, for durability.
- Add a ball-bearing or bushing at the propeller shaft.
- Optimise blade pitch with simple CFD or airflow measurements.
- Add a folding handle and a clip-on lanyard.
- Compare printed gears with a moulded gear set.

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References are from memory and not verified against the sources; check them before submitting.